

Basics 19: Plasma Distributions and Temperature

III - Elementary plasma physics

Velocity-space probability, Maxwellians, moments, anisotropy, and energetic tails

Textbook draft, version 1.0. A kinetic plasma is described by a distribution over position and velocity rather than one trajectory or one bulk velocity. This chapter builds the distribution-function language, derives Maxwellian moments, and clarifies the statistical meaning of plasma temperature. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

Contents

1	Learning objectives	2
2	Prerequisites and reading strategy	2
3	Distribution function and normalization	2
4	One-dimensional Maxwellian	3
5	Three-dimensional Maxwellian and kinetic energy	3
6	Moments and the pressure tensor	3
7	Thermal-speed conventions	4
8	Anisotropic distributions	4
9	Nonthermal and energetic populations	4
10	Particle markers and sampling error	5
11	Computational laboratory	5
12	Summary	5

Symbols and acronyms used in this document	5
13 Exercises	6
14 References	6

1 Learning objectives

After completing this note, the reader should be able to:

- Interpret the distribution function as phase-space density.
- Normalize one- and three-dimensional Maxwellian distributions.
- Derive density, flow velocity, pressure, and temperature moments.
- Distinguish common thermal-speed conventions.
- Represent anisotropic and nonthermal populations.
- Connect particle-marker samples to continuum moments.

2 Prerequisites and reading strategy

- Basics 1, 2, 6, 10, and 17.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Distribution function and normalization

For species s , let $f_s(\mathbf{x}, \mathbf{v}, t)$ satisfy

$$dN_s = f_s d^3x d^3v. \quad (3.1)$$

Its velocity integral is number density:

$$n_s(\mathbf{x}, t) = \int f_s d^3v. \quad (3.2)$$

Some communities normalize f to unity and write nf for phase-space density. The convention must be explicit because it changes every moment formula.

4 One-dimensional Maxwellian

Thermal equilibrium in one velocity component has

$$f(v_x) = n \sqrt{\frac{m}{2\pi k_B T}} \exp \left[-\frac{m(v_x - u_x)^2}{2k_B T} \right]. \quad (4.1)$$

The Gaussian integral ensures $\int f dv_x = n$. Its mean is u_x , and its variance is

$$\langle (v_x - u_x)^2 \rangle = \frac{k_B T}{m}. \quad (4.2)$$

Temperature therefore measures random kinetic-energy spread, not bulk flow speed.

5 Three-dimensional Maxwellian and kinetic energy

For an isotropic equilibrium,

$$f_M(\mathbf{v}) = n \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp \left[-\frac{m|\mathbf{v} - \mathbf{u}|^2}{2k_B T} \right]. \quad (5.1)$$

Each quadratic degree of freedom contributes $k_B T/2$:

$$\frac{m}{2n} \int |\mathbf{v} - \mathbf{u}|^2 f_M d^3v = \frac{3}{2} k_B T. \quad (5.2)$$

Bulk kinetic energy $mnu^2/2$ is separate from internal thermal energy.

6 Moments and the pressure tensor

The bulk velocity is

$$\mathbf{u}_s = \frac{1}{n_s} \int \mathbf{v} f_s d^3v. \quad (6.1)$$

Define peculiar velocity $\mathbf{c} = \mathbf{v} - \mathbf{u}_s$. The pressure tensor is

$$\mathbf{P}_s = m_s \int \mathbf{c} \mathbf{c} f_s d^3v. \quad (6.2)$$

For an isotropic distribution, $\mathbf{P}_s = p_s \mathbf{I}$ and

$$p_s = n_s k_B T_s = \frac{1}{3} \text{tr } \mathbf{P}_s. \quad (6.3)$$

The heat flux is a third-order moment,

$$\mathbf{q}_s = \frac{m_s}{2} \int c^2 \mathbf{c} f_s d^3v. \quad (6.4)$$

7 Thermal-speed conventions

Several definitions are common:

$$v_T = \sqrt{\frac{k_B T}{m}}, \quad v_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}, \quad v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}. \quad (7.1)$$

They are respectively a one-component standard deviation, most-probable speed, and root-mean-square speed. Calling all three “thermal speed” without a definition creates factor-of- $\sqrt{2}$ or $\sqrt{3}$ errors.

8 Anisotropic distributions

Magnetized collisionless plasmas often have different parallel and perpendicular temperatures. A bi-Maxwellian can be written

$$f_b = n \left(\frac{m}{2\pi k_B T_\perp} \right) \left(\frac{m}{2\pi k_B T_\parallel} \right)^{1/2} \exp \left[-\frac{mv_\perp^2}{2k_B T_\perp} - \frac{m(v_\parallel - u_\parallel)^2}{2k_B T_\parallel} \right]. \quad (8.1)$$

Its pressure tensor has $p_\parallel = nk_B T_\parallel$ and $p_\perp = nk_B T_\perp$. Such anisotropy can drive kinetic instabilities.

9 Nonthermal and energetic populations

Fusion alpha particles and neutral-beam ions are not generally Maxwellian. A slowing-down distribution has a characteristic form

$$f_{\text{sd}}(v) \propto \frac{1}{v^3 + v_c^3} \quad (9.1)$$

over a finite birth-energy range. Beams may be strongly anisotropic. A small energetic population can dominate wave drive even when its density is much lower than the thermal-plasma density because resonance depends on phase-space gradients and particle energy.

10 Particle markers and sampling error

A Monte Carlo representation uses weighted markers:

$$f(\mathbf{z}) \approx \sum_{p=1}^{N_p} w_p K_h(\mathbf{z} - \mathbf{z}_p). \quad (10.1)$$

Moments become weighted sums. Statistical error typically decreases as $N_p^{-1/2}$ for independent samples, while interpolation kernels introduce bias and smoothing. Verification should compare sampled moments with analytic distribution moments.

11 Computational laboratory

Sample one- and three-dimensional Maxwellians, reconstruct histograms, and estimate density, flow, pressure, and temperature. Compare thermal-speed definitions. Repeat for a bi-Maxwellian and a simple slowing-down distribution, then quantify Monte Carlo convergence.

12 Summary

- A distribution function describes particle density in phase space.
- Temperature is a moment of random velocity spread, not bulk motion.
- Pressure is generally a tensor and becomes scalar only under isotropy.
- Thermal-speed conventions must be stated explicitly.
- Energetic and anisotropic populations require non-Maxwellian models and careful sampling.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
f_s	distribution function	Species phase-space density.
$\mathbf{u}_s$	bulk velocity	First velocity moment divided by density.
$\mathbf{P}_s$	pressure tensor	Second central velocity moment.
$\mathbf{q}_s$	heat flux	Third central moment.
$\mathbf{c}$	peculiar velocity	$\mathbf{v} - \mathbf{u}_s$.

Symbol / acronym	Name	Meaning in this document
$T_{\parallel}, T_{\perp}$	anisotropic temperatures	Parallel and perpendicular random-energy measures.

13 Exercises

1. Normalize the one-dimensional Maxwellian.
2. Derive $p = nk_B T$ from the isotropic pressure tensor.
3. Separate bulk and thermal kinetic energy for a shifted Maxwellian.
4. Calculate $T_{\parallel}$ and $T_{\perp}$ from sampled velocities.
5. Explain why a low-density energetic tail can strongly affect an Alfvén eigenmode.

14 References

References

- [1] G. B. Arfken, H. J. Weber, and F. E. Harris, *Mathematical Methods for Physicists*, 7th ed., Academic Press (2013).
- [2] G. Strang, *Linear Algebra and Its Applications*, 4th ed., Brooks/Cole (2006).
- [3] W. A. Strauss, *Partial Differential Equations: An Introduction*, 2nd ed., Wiley (2007).
- [4] F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion*, 3rd ed., Springer (2016).
- [5] R. J. Goldston and P. H. Rutherford, *Introduction to Plasma Physics*, CRC Press (1995).
- [6] P. M. Bellan, *Fundamentals of Plasma Physics*, Cambridge University Press (2006).
- [7] T. H. Stix, *Waves in Plasmas*, American Institute of Physics (1992).
- [8] D. A. Gurnett and A. Bhattacharjee, *Introduction to Plasma Physics*, 2nd ed., Cambridge University Press (2017).